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SCHOOL OF INFORMATION SCIENCE

INTERNSHIP REPORT

Intern at Space Science and Geospatial Institute
Duration - Two months
Supervised by Ghion Ashenafi and Bemnet Ayalew

Student Name: Hani Abesha

Student ID: UGR/7363/15

Section: A

Submitted to: Instructor Miftah

Submission Date: November 22, 2025

Student Declaration

I, Hani Abesha, affirm that this report is a product of my original efforts during my internship at Space Science and Geospatial Institute, where I worked in the 3T's (Training and Technology Transfer) department. It accurately represents the experiences, responsibilities, and knowledge acquired throughout my internship. I also certify that this report has not been submitted to any other institution for academic purposes, guaranteeing its originality and authenticity.

Supervisor's Declaration

I, Ghion Ashenafi, hereby affirm that I supervised Hani Abesha during her internship at the Space Science and Geospatial Institute in the 3T's (Training and Technology Transfer) Department. Throughout the program, I closely oversaw the student's activities and noted the dedication and commitment demonstrated in carrying out assigned tasks. The effort and professionalism shown contributed positively to the department and reflected a strong work ethic. I appreciate the valuable contribution made during the internship and hereby extend my approval of the work accomplished under my supervision.

Acknowledgement

First and foremost, I am deeply grateful to God, whose guidance and presence have been with me throughout this journey.

Second, I would like to express my sincere gratitude to the **Space Science and Geospatial Institute** for giving me the opportunity to complete my internship in the **3T's (Training and Technology Transfer) Department**. The experience has been highly valuable and has provided me with practical exposure that has enriched my academic knowledge.

I would also like to acknowledge my advisor, for formally overseeing my internship process and for ensuring that my work was properly evaluated within the academic framework.

My heartfelt appreciation also goes to my **university** for facilitating this internship program and for continuously supporting students in bridging the gap between theory and practice. I am also very grateful to my family for their constant encouragement and support.

Finally, I extend my thanks to all colleagues and staff at the institute who contributed to my learning and made my internship experience productive and meaningful.

Executive Summary

This report provides a detailed account of my internship at **Space Science and Geospatial Institute (SSGI)**, carried out in the **3T's (Training and Technology Transfer) Department**. The internship was an integral part of my academic journey at my university, designed to integrate classroom knowledge with real-world application in a professional setting.

During my two-month internship at the Space Science and Geospatial Institute (SSGI), under the 3T's Department, I had the opportunity to engage in a highly practical and intensive project: the design and development of an **Internship Management System**. Unlike traditional internship experiences that often include rotations and training sessions, my work was focused entirely on building a fully functional system that will be used by the Institute for managing future internship intakes.

The **Internship Management System** was conceived to solve a critical problem: streamlining the process of handling internship applications, department requirements, approvals, and student placements. Traditionally, these processes were managed manually, resulting in inefficiencies, delays, and lack of coordination. The system we built provides an integrated digital solution where departments can submit internship requirements, students can apply online, and administrators can review, and approve applications efficiently.

The workflow of the system is structured as follows: departments begin by submitting requirements, and students apply through an application form. Both inputs are stored in the system's database, after which a matching algorithm aligns student applications with departmental needs. The administrator then reviews and approves matches, generates approval letters, and facilitates student registration. Once registered, the student is officially recognized as an intern and their progress can be monitored through the system.

The project was developed using HTML, CSS, JavaScript, Vanilla and Bootstrap for the frontend, and Python/Django with PostgreSQL for the backend. The development process was intensive, requiring close collaboration, technical problem-solving, and iterative improvements. One of the most challenging but rewarding aspects of the project was requirement gathering. Initially, aligning with stakeholders' expectations was difficult, but this experience highlighted the importance of clear and consistent communication. I came to understand that bridging the gap between technical developers and non-technical stakeholders is as critical as coding itself.

In summary, this internship provided me with practical exposure to full-cycle system development, from requirement gathering to implementation. It deepened my appreciation of teamwork, communication, and the real-world challenges of software engineering. More importantly, it reinforced the idea that the success of any project depends not only on technical expertise but also on the ability to listen, communicate, and deliver solutions that truly address client needs.

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List of Acronyms

SSGI ––––––––––––––––––– Space Science and Geospatial Institute
IMS ––––––––––––––––––– Internship Management System
3T's ––––––––––––––––––– Training and Technology Transfer
UAT ––––––––––––––––––– User Acceptance Testing
HTML –––––––––––––––– Hyper Text Markup Language
CSS ––––––––––––––––––– Cascading Style Sheets
SQL ––––––––––––––––––– Structured Query Language
MySQL –––––––––––––– Relational database management system
PostgreSQL –––––––––– Relational database system
ESSTI –––––––––––––––––Ethiopian Space Science and Technology Institute
GII ––––––––––––––––––– Geospatial Information Institute
GIS ––––––––––––––––––Geographic Information System
UAV –––––––––––––––––Unmanned Aerial Vehicle
EORC –––––––––––––––––Entoto Observatory and Research Center
ESSS –––––––––––––––––Ethiopian Space Science Society
EMA –––––––––––––––––Ethiopian Mapping Agency

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CHAPTER ONE

Project Title: Internship Management System

1.1 Short Summary of the Project

The **Internship Management System** is a web-based application designed to streamline and automate the management of internship processes within the institution. The system facilitates seamless communication between students, departments, and administrative staff by providing an organized platform for internship applications, departmental requirements submissions, approvals, and placements.

Students can submit their internship applications online, while departments can specify and submit their requirements for interns. Administrative users have the ability to review applications, approve student placements, and monitor the overall internship workflow. This system eliminates the need for manual tracking, reduces paperwork, and ensures timely and accurate handling of internship data. Ultimately, it enhances efficiency, transparency, and accessibility for all stakeholders involved in the internship process.

1.2 Problem Statement & Justification

In most institutions, including SSGI, the management of internships is often done manually or through fragmented systems, leading to inefficiencies, delays, and errors. Students face difficulties in finding suitable internship opportunities, submitting their applications, and tracking their application status. Departments struggle to communicate requirements clearly, process multiple applications, and coordinate with administrative staff. Administrators, on the other hand, spend significant time managing paperwork, monitoring progress, and ensuring compliance with internship regulations. What took most of their time and created frustrations in the work environment was that they had to call every applicant to inform them of their application acceptance or rejection. They also had to communicate to departments through phone calls to take their intern requirements.

These challenges result in lost time, miscommunication, and decreased productivity, ultimately affecting the overall internship experience for students and departments alike.

The **justification** for developing an **Internship Management System** lies in its ability to centralize and automate the entire internship workflow. By providing an online platform for application submission, departmental requirement management, and administrative oversight, the system addresses inefficiencies, reduces manual errors, wastage of time and ensures transparency in the internship process. Furthermore, it allows real-time tracking of applications, simplifies coordination between stakeholders, and enhances the overall quality of the internship experience for both students and institutions. It basically reduced their hours of manual work, to be completed in few minutes.

1.3 Objectives of the Project

1.3.1 General Objective

The main objective of the project is to design and implement a fully functional Internship Management System (IMS) for Space Science and Geospatial Institute (SSGI) that streamlines the internship management process for students, departments, and administrative staff.

1.3.2 Specific Objective

1. **Facilitate Student Applications:** Enable students to submit internship applications online, view available opportunities, and track the status of their applications efficiently.
2. **Streamline Department Requirements:** Allow departments to define and submit their internship requirements clearly, making it easier to match students with suitable opportunities.
3. **Matching Algorithm:** the system looks into the student's application, the department's requirement and automatically matches them accordingly.
4. **Enhance Administrative Oversight:** Provide administrators with tools to review applications, student matches, approve placements, generate reports, and monitor the overall internship workflow.
5. **Reduce Manual Processes:** Minimize paperwork, manual follow-ups, and redundant communication by centralizing all internship-related activities in one system.
6. **Improve Transparency and Communication:** Ensure clear communication between students, departments, and administrators, and maintain a transparent record of all internship activities.
7. **Increase Efficiency and Accuracy:** Automate repetitive tasks, reduce errors, and ensure timely handling of applications and approvals.

The system aims to create a more organized, efficient, and user-friendly internship management experience for all stakeholders involved.

1.4 Methodology

The development of the **Internship Management System** followed a structured step-by-step development process inspired by the Agile methodology, to ensure the system was functional, reliable and meets the requirements of students, departments, and administrative staff. This approach allowed us to build, test, and improve the system through feedback at every stage. The methodology adopted includes the following phases:

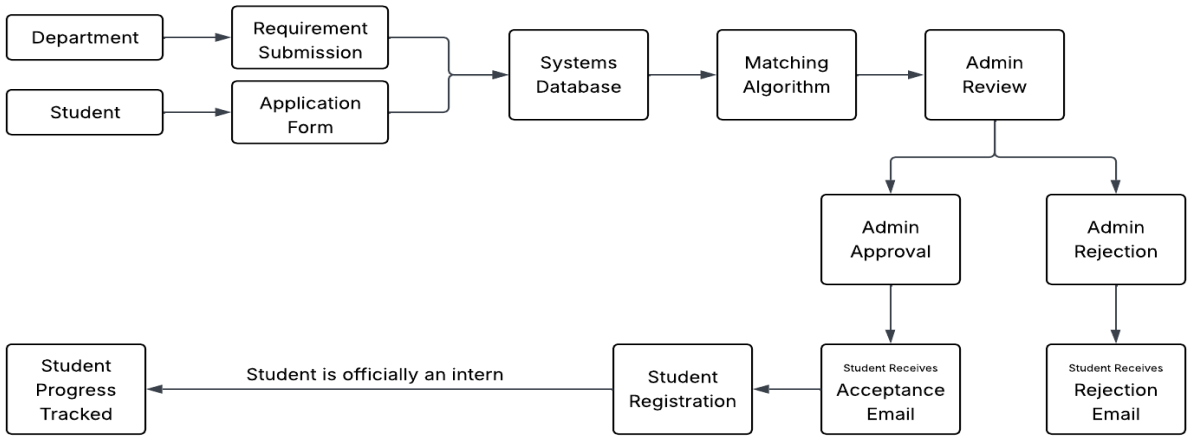
1. **Requirement Analysis:**
    - Conducted a thorough analysis of the internship process in the institution.
    - Identified the key stakeholders, their roles, and their needs.
    - Documented functional requirements (application submission, departmental requirement submission, matching algorithm, approval process) and non-functional requirements (security, usability, ease of use, performance).
2. **System Design:**
    - Designed the architecture of the system using a web-based client-server model.
    - Created data models for students, departments, applications, and approvals.
    - Developed prototypes to visualize processes such as application submission, department's submission and administrator monitoring.
    - Planned the user interface to ensure intuitive navigation for all users.
3. **Development:**
    - Implemented the backend using **Django**, a Python-based web framework, to handle business logic, database interactions, and user authentication.
    - Used **MySQL** as the database to store all system data securely, then later switched to **POSTGESQL** due to changes in stakeholder's requirement.
    - Developed the frontend using HTML, CSS, and JavaScript (Vanilla + Bootstrap) for responsive and user-friendly interfaces.
    - Integrated forms and validation mechanisms to ensure correct data entry by users.
4. **Testing:**
    - Performed **unit testing** to verify individual components of the system.
    - Conducted **integration testing** to ensure that modules work together seamlessly.
    - Carried out **user acceptance testing** (UAT) with sample users to validate system functionality, usability, and performance.
5. **Documentation:**
    - Documented the system's architecture, database schema, and user manual to facilitate future updates and troubleshooting.

The methodology ensured that the project was developed systematically, met the user requirements, and delivered a robust, efficient, and maintainable system for internship management.

1.5 Internship Schedule Table (8 Weeks)

Week	Duration	Activities / Tasks
Week 1	Orientation	Introduction to SSGI, 3T's department overview, understanding project objectives and scope.
Week 2	Requirement Gathering & System Design	Conducted deep requirement gathering with stakeholders; documented requirements; designed system's architecture, workflows, and database schema.
Week 3	Frontend Development	Developed user interfaces and forms; ensured responsiveness and user-friendly navigation.
Week 4	Backend Development	Built core backend functionality using Python/Django; implemented logic for data flow and user requests.
Week 5	Database Integration	Migrated from MySQL to PostgreSQL; structured tables, relationships, and optimized queries.
Week 6	Admin Dashboard Development	Focused on detailed admin dashboard features, enabling system management and matching.
Week 7	Testing & Feedback	Conducted testing sessions; prepared and delivered final presentation to the department and organization; refined features based on feedback.
Week 8	Wrap-Up & Deployment Prep	Finalized project; prepared documentation for deployment; added final refinements and touches for formal deployment.

1.6 Workflow Diagram: Internship Management System



1.7 Literature Review

The management of internships has traditionally been a manual and paper-based process, which is prone to inefficiencies, delays, and errors. Several studies and projects have highlighted the need for digital solutions to streamline internship processes.

1. **Online Internship Management Systems:**
   Previous research shows that web-based internship management systems significantly improve efficiency and transparency. Systems developed in universities often include modules for student applications, departmental requirements, and administrative approvals. For instance, studies have shown that integrating online submission forms and automated notifications reduces delays and enhances communication among stakeholders.
2. **Role of Automation in Internship Processes:**
   Automation of routine tasks such as application tracking, matching, approval workflows, and record maintenance minimizes human error and frees administrative staff to focus on decision-making and quality assurance. Automated systems also provide students with real-time updates on their application status, which improves the overall internship experience.
3. **Database-Driven Solutions:**
   Database management is critical for storing and retrieving large volumes of internship data efficiently. Literature emphasizes the use of relational databases (e.g., MySQL, PostgreSQL) to maintain records of student applications, department requirements, matches and approvals securely. A well-designed database ensures data integrity, scalability, and easy reporting.
4. **User-Centric Design and Usability:**
   Studies on web-based academic systems stress the importance of user-friendly interfaces. Clear navigation, responsive design, and form validation enhance usability for students, departments and administrators alike. Systems with intuitive dashboards and reporting tools are more likely to be adopted and maintained successfully.
5. **Security and Data Privacy Considerations:**
   Protecting sensitive student and department information is essential. Literature recommends implementing authentication, role-based access control, and secure data transmission to safeguard user data. Compliance with data protection standards improves stakeholder trust in the system.

In conclusion, the literature supports the development of a web-based, automated internship management system that integrates database-driven storage, user-friendly interfaces, and secure access. The proposed system aligns with these best practices, providing a modern, efficient, and reliable solution for managing internships within the institution.

CHAPTER TWO

2.1 BRIEF HISTORY OF SPACE SCIENCE AND GEOSPATIAL INSTITUTE

The **Space Science and Geospatial Institute (SSGI)** is a relatively young but strategically important federal institution in Ethiopia. It was officially established in 2022 under Article 57 (1) of Proclamation No. 1263/2021 and defined further by Regulation No. 511/2022 of the Federal Negarit Gazette of the Federal Democratic Republic of Ethiopia. The Institute was formed through the merger of two previously independent organizations: the **Ethiopian Space Science and Technology Institute (ESSTI)** and the **Geospatial Information Institute (GII)**. The unification of these institutions was driven by the government's vision of consolidating national expertise and resources in space science and geospatial technologies under one umbrella, thereby ensuring efficiency, stronger coordination, and enhanced national capacity in these fields.

Although SSGI itself is recent, its roots go back several decades. Ethiopia has a long and rich history of astronomical and geospatial exploration. Traditional knowledge systems included early celestial observations, while the modern era of space science in Ethiopia began in the 1950s, when the country rekindled its scientific interest in astronomy. A milestone came in 2004, with the establishment of the **Ethiopian Space Science Society (ESSS)**, which played a major role in raising public awareness and mobilizing support for national space initiatives.

Building on this momentum, Ethiopia launched the **Entoto Observatory and Research Center (EORC)** in 2013, the first observatory of its kind in East Africa. Soon after, the **Ethiopian Space Science and Technology Institute (ESSTI)** was created in **2016** to institutionalize the country's space science research, satellite development, and policy implementation. Parallel to this, Ethiopia had been advancing in the field of geospatial technologies since the mid-20th century. The journey began with the establishment of the **Ethiopian Cartographic Mapping and Geography Institute in 1954**, which later evolved into the **Ethiopian Mapping Agency (EMA)** and then into the **Geospatial Information Institute (GII)**, reflecting the rapid advancements in mapping and geoinformatics.

By merging **ESSTI** and **GII** in **2022**, the Ethiopian government formally created the Space Science and Geospatial Institute (SSGI). This merger symbolized not only an administrative consolidation but also a strategic national commitment to leverage space and geospatial science for Ethiopia's socio-economic development, national security, environmental management, and digital transformation.

Today, SSGI continues to expand its role as a hub for research, training, consultancy, and technology development in both the space and geospatial domains. It represents Ethiopia in regional and international platforms, aiming to make the country a competitive force in Africa's space and geospatial sector.

FIGURE 1 and FIGURE 2

2.2 Main Products and Services

The **Space Science and Geospatial Institute (SSGI)** provide a wide range of products and services that support Ethiopia's research, innovation, and decision-making. Its outputs benefit government institutions, industries, and society at large, while advancing the country's digital economy and sustainable development goals.

1. **Space Science and Satellite Services**
    - Development, operation, and utilization of satellites.
    - Astronomical research and astrophysical observation.
    - Implementation of national space science policy and strategy.
2. **Geospatial Information Services**
    - Mapping, cartography, and geospatial database management.
    - Remote sensing, GIS platforms, and spatial analysis.
    - Generation of digital maps, atlases, and earth observation products.
3. **Research, Innovation, and Technology Transfer**
    - Applied and theoretical research in space and geospatial sciences.
    - Development of new technologies and transfer to industries and agencies.
    - Academic, scientific, and international collaborations.
4. **Training and Capacity Building**
    - Specialized training in satellite technology, GIS, and remote sensing.
    - Professional consultancy, technical support, and advisory services.
    - Awareness creation, outreach programs, and postgraduate capacity development.
5. **National and Community Services**
    - Disaster management, food security monitoring, and environmental protection.
    - Digital addressing systems and national integration through shared geospatial platforms.
    - Representation of Ethiopia in regional and international space forums.

SSGI's services go beyond research, extending to agriculture, urban planning, disaster management, education, and national security, making it vital for Ethiopia's digital economy and sustainable development goals.

2.3 Main Customers and End Users of SSGI's Products and Services

The **Space Science and Geospatial Institute (SSGI)** serve a wide range of stakeholders, ensuring its products and services have both **scientific value** and **socio-economic impact**. Its work spans from *underground → earth → to space*, supporting decision-making and national development.

1. **Government Ministries and Agencies**
    - The **Ministry of Innovation and Technology (MinT)** is the primary partner, guiding priorities and oversight.
    - Other key users include the Ministries of Agriculture, Urban and Infrastructure Development, Water and Energy, and the National Disaster Risk Management Commission, all relying on satellite and geospatial data for planning, monitoring, and decision-making.
2. **Regional and Local Governments**
    - Utilize SSGI's maps, spatial datasets, and geospatial services for urban zoning, infrastructure planning, land management, and resource allocation.
3. **Educational and Research Institutions**
    - Universities and research centers benefit from access to scientific data, collaborative research, training, and postgraduate capacity-building programs.
4. **Private Sector and Industry**
    - Telecommunications, construction, engineering, mining, and agribusiness companies use SSGI's geospatial and satellite data for network planning, precision farming, land development, and supply chain optimization.
5. **International Organizations and Partners**
    - Collaborate with SSGI on **satellite development, capacity building, and technology transfer**, strengthening Ethiopia's global presence in space and geospatial science.
6. **General Public and Communities**
    - Citizens benefit indirectly through improved food security, disaster preparedness, digital addressing systems, and public awareness programs.
7. **Policy Makers and Decision Makers**
    - Leaders at national, regional, and local levels rely on SSGI's outputs for economic planning, climate adaptation strategies, and sustainable resource management.

Through its **Enabling Sector**, SSGI ensures that research outputs and geospatial data are efficiently linked with end-users across all these categories, acting as a vital bridge between science, technology, and sustainable development in Ethiopia.

2.4 Organizational Structure of SSGI

The **Space Science and Geospatial Institute (SSGI)** is an autonomous federal institute under the **Ministry of Innovation and Technology (MinT)**. Its organizational structure reflects its broad mandate of working *"from underground → earth → to space."*

High-Level Management

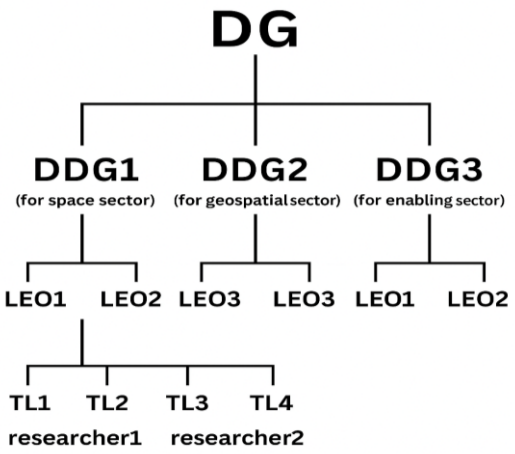
- **Director General (DG)** – The chief executive of the institute, responsible for overall leadership and strategic direction.
- **Deputy Director Generals (DDG1, DDG2, DDG3)** – Three deputies who oversee core functions of the institute under the DG.
- **Lead Executive Officers (LEOs)** – Each DDG supervises Lead Executive Officers who coordinate departments and projects.
- **Team Leaders (TLs)** – LEOs are supported by Team Leaders who manage day-to-day research and operations.
- **Researchers & Staff** – The foundation of the organization, carrying out scientific, geospatial, and administrative tasks.

Core Sectors

1. **Space Sector** – Focuses on aerospace engineering, space science, astronomy, astrophysics, and satellite operations.
2. **Geospatial Sector** – Handles geospatial data generation, manipulation, mapping, and infrastructure management.
3. **Enabling Sector** – Provides cross-cutting support, including system administration, IT services, and linking data with external stakeholders.

Main Wings

- **Scientific Wing (Core Office)** – Dedicated to research, innovation, and scientific operations.
- **Administrative/Supportive Wing** – Responsible for HR, finance, logistics, and general support services.



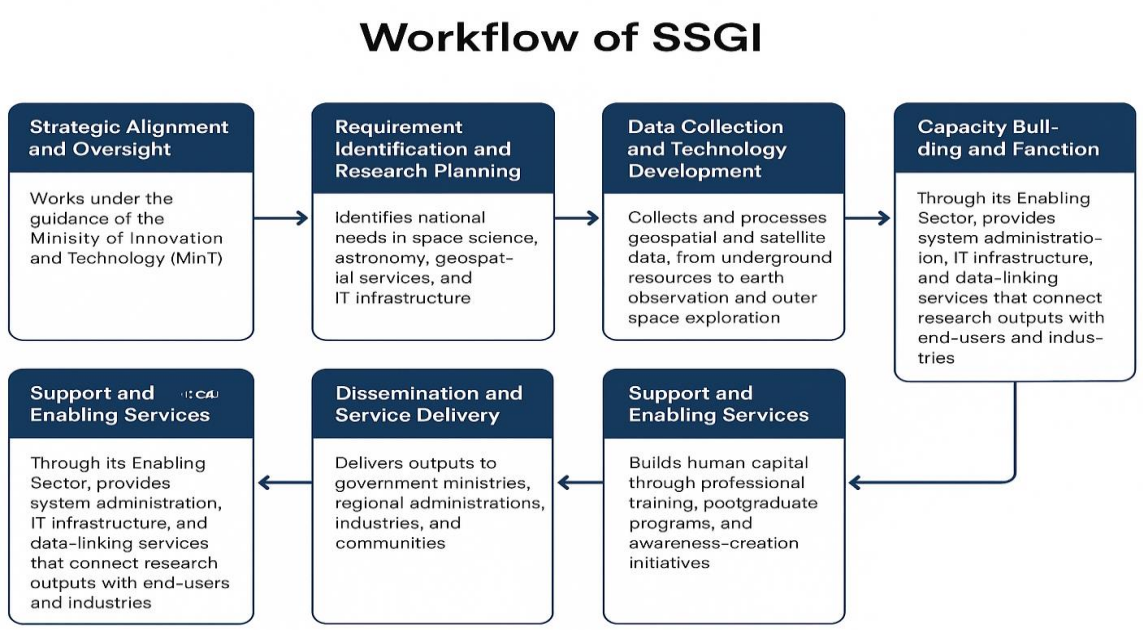
2.5 Workflow of SSGI

The workflow of the **Space Science and Geospatial Institute (SSGI)** is designed to integrate research, technology development, and service delivery in alignment with Ethiopia's national development priorities. Its operations typically follow these stages:

1. **Strategic Alignment and Oversight**
    - Works under the guidance of the **Ministry of Innovation and Technology (MinT)** to align projects with Ethiopia's development plans and digital economy strategies.
    - Ensures activities reflect national priorities in science, geospatial services, and space technology.
2. **Requirement Identification and Research Planning**
    - Identifies national needs in space science, astronomy, geospatial services, and IT infrastructure.
    - Plans research programs, satellite initiatives, and geospatial mapping projects.
3. **Data Collection and Technology Development**
    - Collects and processes geospatial and satellite data, from underground resources to earth observation and outer space exploration.
    - Develops technologies such as satellites, UAVs, and GIS platforms to support data generation.
4. **Analysis and Product Generation**
    - Transforms raw data into usable outputs such as maps, atlases, models, spatial datasets, and scientific reports.

- Conducts scientific studies that support decision-making in areas like agriculture, urban planning, and climate monitoring.

5. **Dissemination and Service Delivery**
    - Delivers outputs to government ministries, regional administrations, industries, and communities.
    - Provides consultancy, advisory services, and technical support to stakeholders.
6. **Capacity Building and Collaboration**
    - Builds human capital through professional training, postgraduate programs, and awareness-creation initiatives.
    - Engages with international organizations, universities, and industries for partnerships and technology transfer.
7. **Support and Enabling Services**
    - Through its **Enabling Sector**, provides system administration, IT infrastructure, and data-linking services that connect research outputs with end-users and industries.
8. **Monitoring and Feedback**
    - Evaluates the impact of its products and services.
    - Incorporates feedback to refine research activities, tools, and technology platforms.



CHAPTER THREE

OVERALL INTERNSHIP EXPERIENCE

3.1 WHY I CHOSE THIS COMPANY

I chose to intern at the **Space Science and Geospatial Institute (SSGI)** because of its exceptional professionalism, strong work ethic, and highly organized structure. It is the only space science–related organization IN THE COUNTRY, which immediately drew my attention as a unique opportunity to gain experience in a specialized field. The institute's environment is incredibly clean, quiet, and high-standard—it almost resembles a hospital with specialist surgeons only, if you looked at the images (*Figure 3*, *Figure 4*, *Figure 5* *and* *Figure 6*). Every individual there has a clear purpose, and the overall atmosphere reflects dedication and focus.

Being a relatively new organization, I saw tremendous potential for personal and professional growth. I believed this internship would allow me to showcase my abilities while learning and expanding my potential. Honestly, I did not expect to be selected at all, so receiving the acceptance call was both surprising and exciting. In fact, at the time, I was attending another interview, and although I was also accepted there, I chose SSGI because it aligned perfectly with what I sought in an internship experience.

Working at SSGI has been a true honor. I genuinely looked forward to going to work every morning, and the experience has been both inspiring and motivating.

FIGURE 3

FIGURE 4

3.2 Which Section of the Company I Worked In and Why?

During my internship at SSGI, I was assigned to the **3T's Department (Training and Technology Transfer)**. This department plays a crucial role in developing and managing web-based application projects, making it an ideal place for interns with a background in computer science or information systems, especially those who have experience in both frontend and backend development.

What made 3T's particularly exciting for me was its involvement in the internship program itself. The department handles the acceptance and rejection of internship applications, as internships are considered an integral part of the institute's training initiatives. Because of this, the project I worked on—an Internship Management System—was directly aligned with the department's operations. This gave me a unique opportunity to contribute something meaningful that could immediately impact the department's workflow.

I accepted my placement to this section because it allowed me to combine my technical skills with a real organizational need, and also gave me insight into how a specialized research institute manages its training programs. Working in 3T not only challenged me to apply what I had learned in my coursework but also gave me the satisfaction of knowing that my work would be genuinely useful to the department.

FIGURE 5

FIGURE 6

3.3 WORKFLOW IN 3T'S DEPARTMENT

The **Training and Technology Transfer (3T's) Department** plays a central role in building human capacity, facilitating technology transfer, and supporting innovation within the Space Science and Geospatial Institute (SSGI). However, as a relatively new department, its workflow is not yet fully standardized and is often shaped by the nature of the task and the departments involved.

According to the Lead Executive Officer (LEO) of 3T's, the department has a **"complicated relationship with other departments,"** since its work overlaps with multiple sectors of SSGI. For example, when organizing postgraduate training, 3T's must coordinate with the Scientific Wing to select trainers, with Administration to manage logistics, and with MinT or universities for academic oversight. Similarly, in technology transfer activities, it collaborates closely with technical teams in the Space and Geospatial sectors to identify innovations, and with external stakeholders to facilitate outreach.

Because of these cross-cutting responsibilities, the workflow currently varies case by case:

- Some activities are initiated internally (e.g., staff training or awareness programs).
- Others are externally driven (e.g., requests for consultancy, startup incubation, or technology transfer).
- In both cases, 3T's acts as a bridge between SSGI's technical departments, administrative units, and outside organizations.

At present, the department is in the process of developing a more structured workflow model. This includes:

1. Establishing clearer communication channels with other departments.
2. Defining step-by-step processes for training, outreach, and technology transfer activities.
3. Creating documentation and reporting systems to track performance and impact.

3.4 INTERNSHIP TASKS

During my internship at the 3T's Department, my main responsibility revolved around developing and enhancing the Internship Management System, a project directly aligned with the department's operations. My work was both technical and highly impactful, allowing me to gain hands-on experience while contributing to the institute's efficiency.

Some of the key tasks I performed include:

1. **Deep Requirement Gathering:** I engaged closely with department staff to understand their needs, workflows, and challenges. This involved interviews, observation, and detailed analysis of existing processes to ensure the system would address real problems effectively.
2. **System Development and Integration:** I worked on both frontend and backend, mostly backend development, creating user-friendly interfaces for students to submit applications, for department to submit their requirements and for admins to review and approve them. I designed pages, dashboards, and notification systems to make the workflow seamless and intuitive. See Appendix B for screenshots of the system interface.

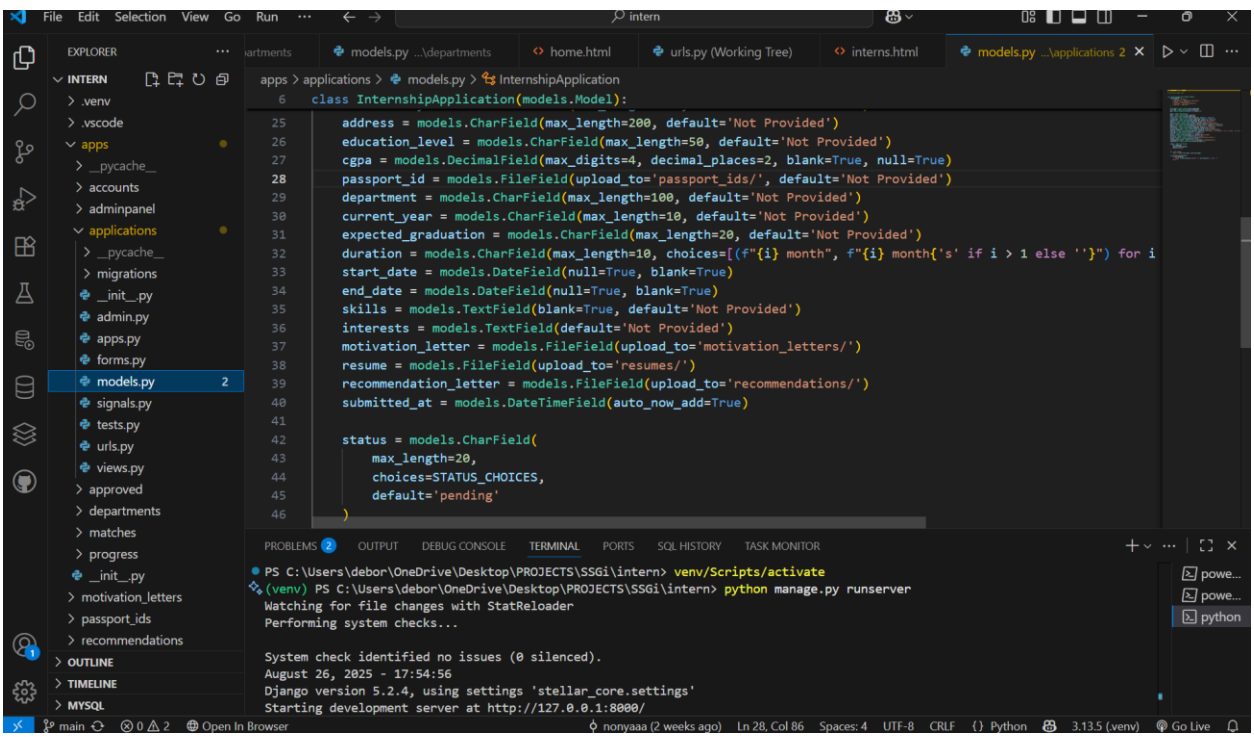


FIGURE 7

3. **Database Management:** I managed the central database, ensuring that student applications, department requirements, and approval records were accurately stored and easily retrievable. I implemented proper relationships, validations, and indexing to improve system efficiency.

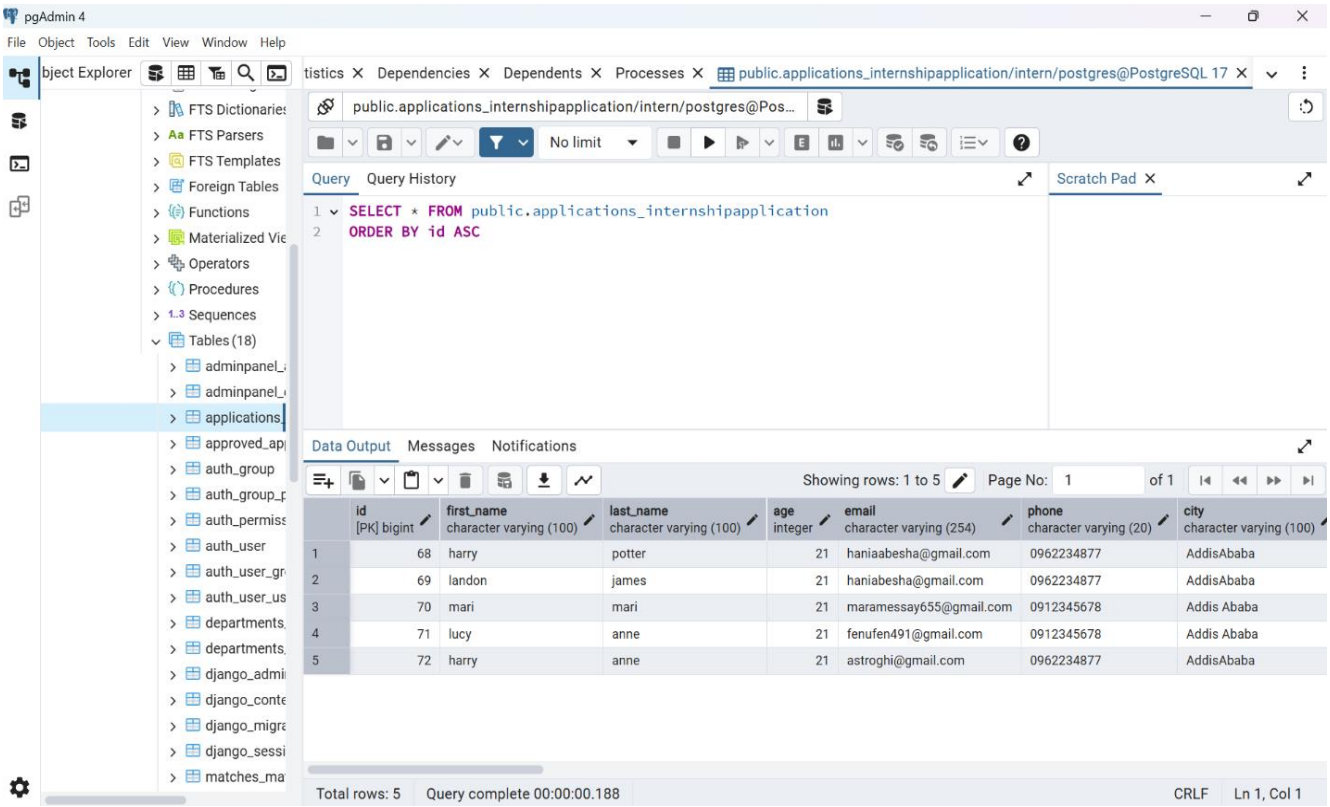


FIGURE 8

4. **Matching Algorithm Implementation:** I contributed to coding the matching algorithm, which aligns students with departments based on department's requirement and student data. This required logical thinking, attention to detail, and iterative testing to ensure fair and accurate matches. This was the most important and the most desired feature for our department. It saves a lot of time, energy, and paperwork for future internship recruitment.

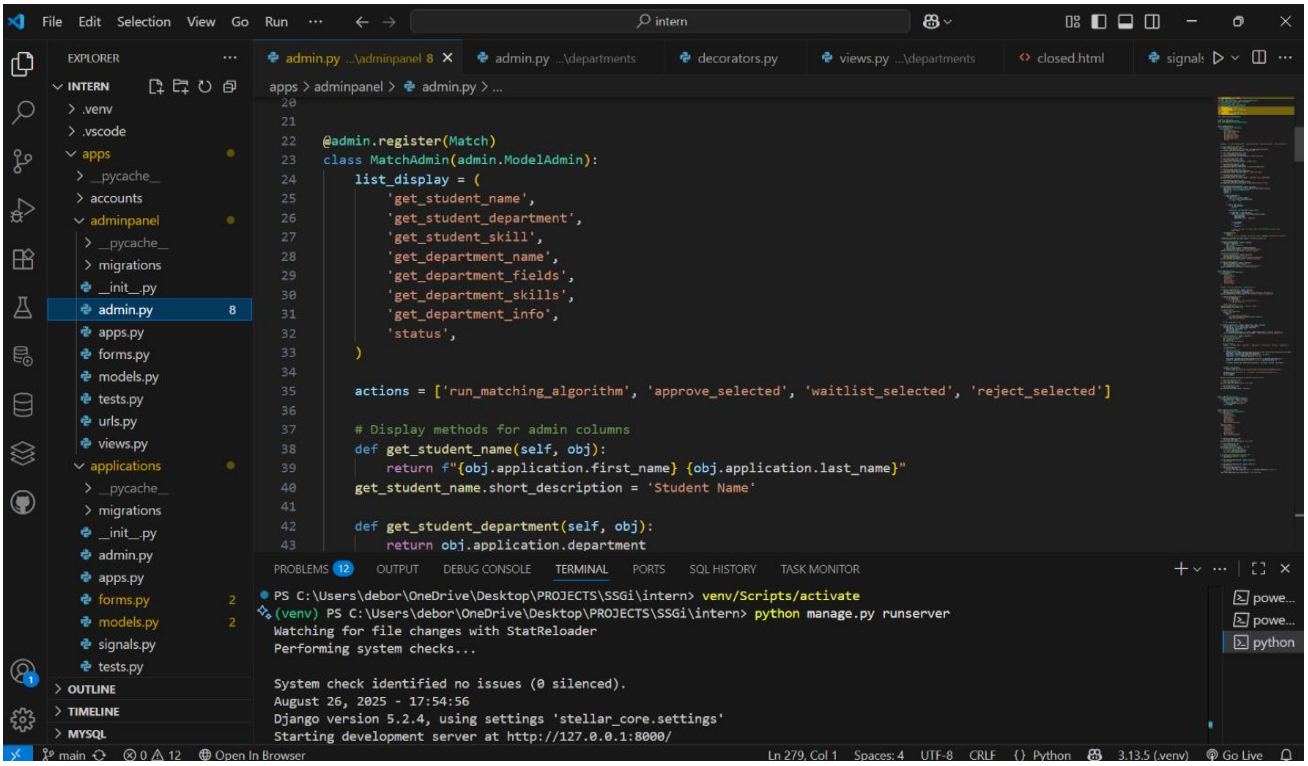


FIGURE 9

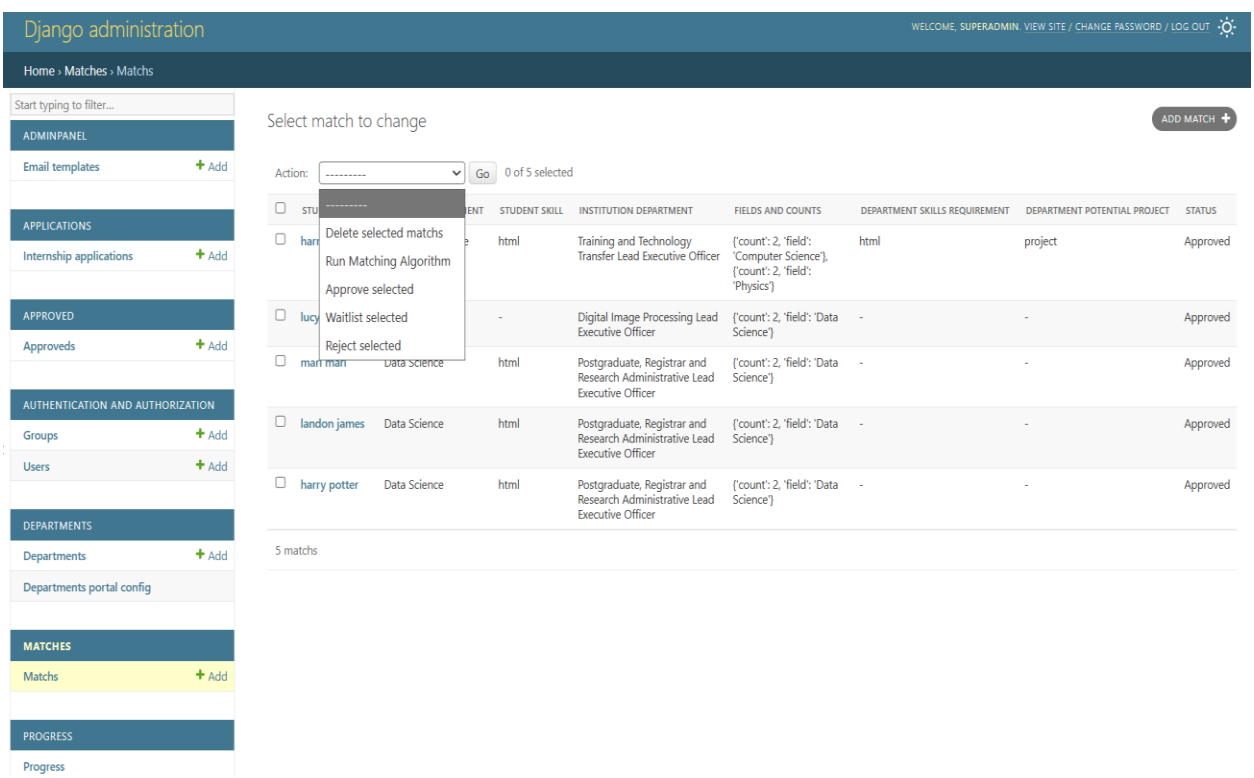


FIGURE 10

5. **Testing and Debugging:** I rigorously tested the system to identify and fix bugs, ensuring that all functionalities—submission, review, approval, and notifications—worked flawlessly.
6. **Weekly Reports and Progress Presentations:** To maintain clear communication with the department, we had weekly reports detailing completed tasks, ongoing challenges, and next steps. I also delivered weekly progress presentations, receiving valuable feedback that guided further improvements to the system.

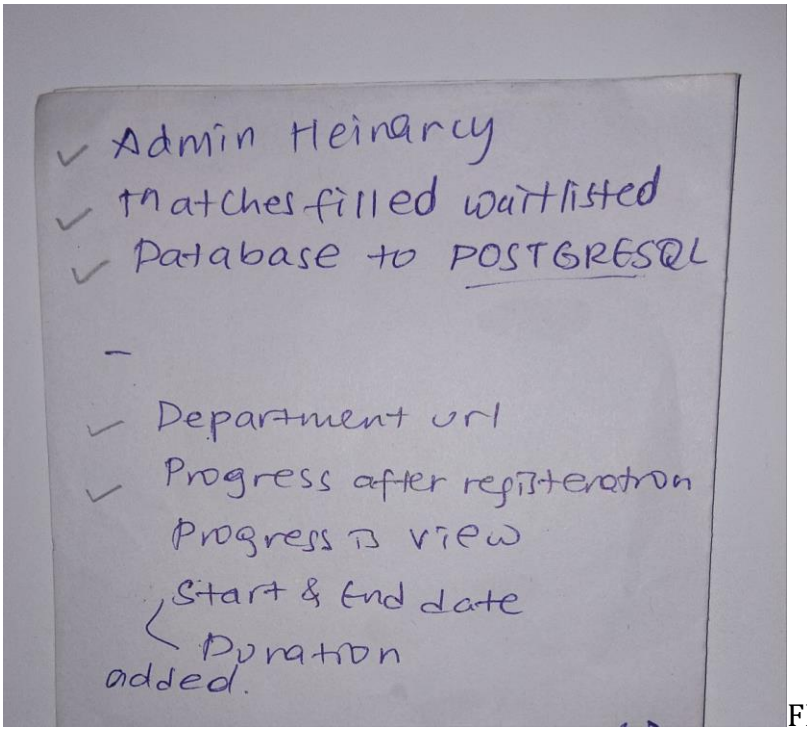


FIGURE 11 and FIGURE 12

7. **Documentation and Reporting:** I prepared detailed documentation of the system's architecture, features, and workflow. This helped the department (the 3T's) to understand the system better and provided a reference for future interns or developers.
8. **User Support and Feedback Integration:** I collected feedback from both students and department staff, identifying areas for improvement and implementing adjustments to make the system more intuitive and efficient.

Through these tasks, I was not just coding—I was analyzing problems, proposing solutions, streamlining processes, and creating a tool that will have lasting value for the department. Each task reinforced my technical skills while giving me a sense of purpose and tangible impact within a professional environment.

3.5 Mechanical Engineering Methods, Tools, and Techniques I Have Been Using While Performing My Work Tasks

Although my internship was primarily centered on software development, many of the methods, tools, and techniques I used reflected engineering principles that are widely applied in the field of Mechanical Engineering. These approaches helped me work systematically, solve problems effectively, and deliver a reliable solution.

1. **Requirement Analysis and System Design (Analogous to Engineering Design Process):**
   Just as mechanical engineers begin with understanding the specifications of a machine or component, I started by conducting deep requirement gathering. I studied the department's workflow, interviewed staff, and analyzed their challenges before designing the Internship Management System. This mirrors the design and prototyping phase in mechanical engineering.
2. **Workflow Modeling and Process Mapping (Equivalent to Process Engineering):**
   We used system diagrams, and structured process mapping to visualize how data should move through the system—from student applications to approval. This is similar to how mechanical engineers model workflows in manufacturing or system operations.
3. **Iterative Prototyping and Testing (Parallel to Product Development):**
   Instead of building the system all at once, I followed an **iterative approach**—designing, testing, debugging, and refining the application in cycles. This is comparable to mechanical engineering prototyping, where models are tested and improved until they meet performance standards.
4. **Precision and Accuracy (Inspired by Measurement and Tolerance Techniques):**

Mechanical engineers emphasize accuracy in design and measurements. I applied the same principle in database design, algorithm development, and validation checks, ensuring that data was stored correctly, matches were precise, and system outputs were reliable.

5. **Problem-Solving Methodology (Root Cause Analysis):**
   Whenever bugs, mismatches, or system inefficiencies occurred, I used a root cause analysis approach, similar to how mechanical engineers troubleshoot machinery issues. This structured problem-solving method helped me resolve errors effectively.
6. **Tools and Technologies Used:**
   - **Software Tools:** Django (backend framework), HTML/CSS/JavaScript (frontend), POSTGRESQL (database), Git/GitHub (version control).
   - **Documentation Tools:** System design, technical reports, and weekly progress presentations—mirroring how engineers document designs, test results, and maintenance logs.
   - **Communication Tools:** Weekly reports and presentations acted as feedback loops, similar to engineering progress reviews.
7. **Continuous Improvement (Lean Engineering Principle):**
   By collecting feedback from users and staff, I refined the system multiple times to make it more efficient. This reflects the Kaizen/continuous improvement approach often used in engineering to optimize processes over time.

In summary, while the specific tools were digital rather than mechanical, the engineering mindset—structured design, precision, iterative testing, and continuous improvement—guided every step of my work. This reinforced the fact that engineering principles are universal, whether applied to building machines or developing software systems.

3.6 Major Challenges and Problems I Faced

Like any meaningful professional experience, my internship at SSGI came with its own set of challenges. While these obstacles were at times frustrating, they ultimately became opportunities for growth, patience, and valuable learning.

1. **Unclear and Constantly Changing Requirements**
   The biggest challenge I faced was working with stakeholders—our department supervisors, and other non-technical members—who often **did not have a clear vision of what they wanted from the system**. Each week during progress presentations, new requirements were added, or previously agreed-upon decisions were reopened for debate. Sometimes they even requested features that were not technically feasible, which created tension and confusion.

This issue often pulled us backward. Instead of moving forward and adding innovative features that we, as technical designers, had planned, we were forced to undo previous work and start over. One memorable example was when, after significant effort spent developing the database in **MySQL**, we were suddenly asked to migrate everything to **PostgreSQL**, requiring us to self-learn a new database system in the middle of the project. It was extremely frustrating and time-consuming.

The experience reminded me of something my lecturer once told us in class: *"Gathering requirements and aligning technical designers with non-technical stakeholders is the hardest part of this job. Customers don't really know what they want."* I felt the truth of that statement during this internship. At times, it even made my team question our future plans and career goals. It also gave me perspective on why many people prefer to run their own businesses—because sometimes decisions made by supervisors or higher-ups can be very difficult to understand from the ground level.

2. **Teamwork and Different Personalities**
   Another challenge I faced was adjusting to the different behaviors and personalities of my teammates. Some were cooperative, enthusiastic, and easy to work with, and from them I built strong friendships. Others were more difficult, with different attitudes and working styles that sometimes clashed with mine. While this created tension occasionally, it also taught me patience, adaptability, and the true meaning of teamwork and collaboration. Learning to communicate effectively and work toward a common goal despite differences was one of the most valuable takeaways from this experience.

3. **Logistical and Practical Challenges**
   There were also smaller, practical challenges that impacted daily life. Our office was located on the sixth floor of the building, with no elevator, which meant climbing up and down several flights of stairs every day. While I sometimes joked about it being good exercise, it was still an inconvenience. Additionally, the institute's location in the middle of a busy street made transportation challenging. We often had to walk nearly half an hour to reach the nearest transport station. While this too became part of the routine, it added an extra layer of fatigue to already demanding days. But, I also loved having my half an hour daily walk, since the area was very greenery and beautiful, it was enjoyable as well.

4. **Additional Issues**
   Beyond the major challenges, there were also smaller but persistent issues such as occasional miscommunications with supervisors and the difficulty of balancing multiple tasks at once. None of these were deal-breakers, but they did add to the complexity of the experience.

However, overcoming these difficulties taught me one of the most important lessons of my internship: the value of strong communication skills. I came to realize that communication is not just a useful skill—it is the most important feature needed in any workplace. Clear,

respectful, and consistent communication can prevent misunderstandings, align expectations, and build stronger collaboration within teams.

Despite the challenges, each experience shaped me into a more resilient and adaptable professional. They gave me practical lessons that no classroom could teach—especially about stakeholder management, patience, teamwork, and above all, communication**.**

3.7 Measures I Took to Overcome Challenges

Facing challenges during my internship was inevitable, but rather than being discouraged, I tried to approach each problem with solutions and practical adjustments. These measures not only helped me navigate difficult situations but also strengthened my professional skills.

1. **Dealing with Unclear and Changing Requirements**
   To handle the constant changes in requirements from stakeholders, I emphasized clear documentation and structured communication. After every meeting or progress presentation, I prepared written notes and confirmations of what was agreed upon. This reduced confusion and served as a reference point when disagreements reappeared later. I also learned to ask precise questions to uncover hidden requirements early on, instead of waiting for them to surface later in the project.
2. **Bridging the Gap Between Technical and Non-Technical Stakeholders**
   One of the biggest lessons was learning how to translate technical language into simple terms that non-technical supervisors and stakeholders could understand. By simplifying explanations and using prototypes, I made it easier for them to see what was being built and why certain requests were not feasible. This approach reduced misunderstandings and helped align expectations.
3. **Adapting to Sudden Technical Changes**
   When we were asked to switch from **MySQL to PostgreSQL** in the middle of the project, I treated it as a learning opportunity. Even though it was frustrating, I spent time researching and experimenting with PostgreSQL until I became comfortable. This mindset of adaptability and continuous learning allowed me to turn a setback into a valuable new skill.
4. **Improving Team Collaboration**
   Working with diverse personalities in my team taught me that patience and empathy are key. I focused on building friendships where possible, but in difficult situations, I relied on professionalism and open dialogue. I encouraged group discussions to share opinions, divide tasks clearly, and set common goals. This made teamwork smoother and less stressful.

5. **Handling Practical and Logistical Issues**
   While climbing six flights of stairs daily and walking long distances to catch transport were not major problems, I reframed them as opportunities for physical activity and discipline. By keeping a positive mindset, I turned small inconveniences into something that actually benefited me in the long run.
6. **Strengthening Communication as a Core Skill**
   Above all, I made a conscious effort to improve my communication skills—whether through weekly reports, progress presentations, or one-on-one conversations with supervisors. I realized that effective communication can prevent many workplace issues before they even occur. This became my most important takeaway and a skill I will continue to practice in my career.

3.8 Results and Discussion

The outcome of my internship project, the **Internship Management System**, was highly successful and rewarding. After months of requirement gathering, design, development, and continuous improvement, we delivered a fully functional system that fulfilled almost every requirement outlined by the 3T Department.

The system is not just a prototype or a class project—it is a real solution intended for actual use within the institute. At the time of completing my internship, the project was in the process of deployment, with department staff preparing to adopt it as part of their official workflow. This gave me an immense sense of pride, knowing that the work my team and I produced would have a direct, tangible impact on the organization.

Some key outcomes include:

- **Automated Workflow:** The system successfully digitized the end-to-end process of internship applications, departmental requirements, and approvals, replacing manual handling with a faster, more efficient solution.
- **Improved Accuracy and Transparency:** By integrating a centralized database and automated matching, the system reduced the chance of errors and ensured that decisions were based on clear, objective data.
- **Ease of Use:** The user-friendly interface made it simple for both students and staff to navigate the system, reducing confusion and training needs.
- **Scalability:** The system was designed in such a way that new features or departments can be added in the future without disrupting the existing functionality.

From a personal perspective, this project was an eye-opener. It taught me how challenging yet rewarding it is to align technical design with real-world organizational needs. While not

every requirement was easy to meet, the end product demonstrated that persistence, teamwork, and adaptability lead to success.

Overall, this project did more than just fulfill an academic requirement—it created something of lasting value for SSGI. The fact that it is being deployed for actual organizational use stands as proof of its effectiveness and the hard work my team put into it.

CHAPTER FOUR

OVERALL BENEFITS FROM MY INTERNSHIP

4.1 Improving My Practical Skills During the Internship

During my internship at the Space Science and Geospatial Institute (SSGI), one of my greatest achievements was the significant improvement of my practical skills. The opportunity to work on the **Internship Management System**, a project designed for real deployment, pushed me to apply classroom knowledge in a professional context while also learning entirely new technologies.

A key area of growth was my ability to self-learn. The project required me to work with Python/Django and PostgreSQL, technologies that were new to me. Although I had only a basic foundation in programming, I quickly familiarized myself with Django's structure—models, views, and templates—as well as PostgreSQL's data handling and integration. This experience gave me confidence that I could independently learn and apply new tools within tight deadlines, a skill that will serve me well in any future role.

I also strengthened my skills in requirement gathering and analysis. Since the system was being built for actual departmental use, it was vital to understand stakeholder needs clearly. I learned how to ask targeted questions, translate user expectations into technical features, and adjust when requirements evolved. Presenting weekly progress reports further sharpened this skill, as I had to explain complex technical concepts in simple, practical terms that both technical and non-technical staff could understand.

Another area of growth was in workflow and documentation. I learned the importance of version control, systematic reporting, and clear documentation for ensuring consistency and long-term usability of the system. These practices not only improved the quality of our work but also taught me how professional software development projects are structured and maintained.

Finally, the internship enhanced my problem-solving skills under real conditions. Whether it was debugging database issues or handling unexpected requirement changes, I learned to approach problems calmly, strategically, and collaboratively. This experience made me

realize that practical problem-solving is not only technical but also about communication, teamwork, and adaptability.

Overall, the internship transformed me from a student with mostly theoretical knowledge into someone capable of applying practical skills in real-world contexts. By self-learning Django and PostgreSQL, conducting requirement analysis, delivering weekly presentations, and solving real problems, I became more confident, resourceful, and prepared for professional software development.

4.2 Upgrading My Theoretical Knowledge During the Internship

Working on the Internship Management System allowed me to see how concepts from my studies—such as software architecture, database management, and system design—are applied in a professional setting.

I strengthened my understanding of software development frameworks, particularly Python/Django, and relational database principles through PostgreSQL. Learning to design models, structure queries, and implement efficient backend processes reinforced the theoretical knowledge I had gained in coursework. The challenges of translating stakeholder requirements into functional system features helped me apply systems analysis and design concepts, showing how theory underpins practical decision-making.

Additionally, preparing weekly progress reports and presentations required me to think critically about system workflows, data management, and logical organization. This not only reinforced my knowledge of project planning, modular design, and algorithmic thinking but also demonstrated the importance of aligning theory with user needs.

The internship deepened my comprehension of theoretical principles by giving me hands-on experience in applying them to a fully functional system, bridging the gap between study and professional practice.

4.3 Enhancing My Industrial Problem-Solving Capability

My internship at SSGI significantly enhanced my ability to solve practical, real-world problems in an industrial setting. Unlike academic exercises, which usually have clear instructions and defined outcomes, the challenges at 3T's Department were dynamic, complex, and sometimes unpredictable.

Working on the Internship Management System, I frequently faced situations where stakeholder requirements were unclear, changing, or technically challenging. This required me to analyze problems critically, identify constraints, and propose feasible solutions. For example, when the project database had to be migrated mid-development from MySQL to PostgreSQL, I quickly learned the new system and ensured the workflow remained

functional. This experience taught me the importance of adaptability and practical problem-solving under pressure.

I also learned how to balance technical possibilities with organizational needs, translating stakeholder requests into actionable tasks while maintaining system integrity. Iteratively testing, debugging, and refining the system improved my decision-making and helped me understand the value of systematic approaches to complex problems.

Moreover, presenting weekly progress updates provided opportunities to address issues collaboratively with both technical and non-technical staff. This reinforced my ability to communicate problems clearly, propose solutions, and align team efforts, which are essential skills in any industrial environment.

This experience honed my industrial problem-solving capability, teaching me to approach challenges with analytical thinking, flexibility, and practical strategies that deliver results in real-world professional settings.

4.4 Developing My Team Playing Skills

During my internship at SSGI, working closely with a team of interns and supervisors in the 3T's Department greatly enhanced my team-playing skills. Collaborating on the Internship Management System project required constant coordination, communication, and adaptability, as each team member had different strengths, working styles, and perspectives.

I learned to listen actively and respect diverse opinions, recognizing that every team member contributes valuable insights. At times, conflicting ideas or approaches led to discussions that were challenging, but they ultimately strengthened our collaboration and improved the system. This taught me patience and the importance of compromise in achieving shared goals.

The internship also improved my task coordination and role management. By clearly dividing responsibilities—such as frontend, backend, routes—we ensured that progress was steady and efficient. Regular weekly meetings kept the team aligned and encouraged open discussion of challenges, which reinforced accountability and collective problem-solving.

Furthermore, I learned how to support teammates during difficult phases, whether by sharing knowledge about Django and PostgreSQL, assisting with debugging, or providing moral encouragement when deadlines were tight. These experiences highlighted that teamwork is not just about completing tasks but also about building trust, motivation, and a cooperative environment.

4.5 Enhancing My Leadership Skills

During my internship at SSGI, I had the opportunity to develop my leadership skills in a real-world, collaborative environment. Working on the Internship Management System project helped me practice the teaching that leadership is not about enforcing one's own way but about understanding the needs and motivations of the team and guiding them effectively.

I learned the importance of listening deeply to everyone's perspectives and allowing room for mistakes. Rather than trying to control every decision, I discovered that being flexible and supportive encourages creativity and problem-solving. Leadership, I realized, is about leading through empathy and guidance, helping team members feel confident, motivated, and satisfied while achieving project goals. In essence, leadership is about leading emotions as much as tasks—creating an environment where people are happy and productive simultaneously.

This experience also improved my ability to coordinate tasks, delegate responsibilities, and encourage collaboration. By balancing direction with autonomy, I learned how to maintain accountability without stifling initiative.

As a result, this internship helped me understand that effective leadership is a combination of empathy, flexibility, and motivation. It is not just about directing work, but about inspiring and guiding people to perform their best while fostering a positive and supportive environment.

4.6 Understanding Work Ethics, Industrial Psychology, and Related Issues

My internship at SSGI provided me with valuable insights into professional work ethics, organizational behavior, and industrial psychology. Observing and participating in the daily workflow of the 3T's Department allowed me to understand how professionalism, discipline, and interpersonal dynamics shape the success of an organization.

I learned the importance of consistency and accountability. Every team member had clear responsibilities and a strong sense of purpose, which reinforced the value of taking ownership of one's tasks. Witnessing how deadlines, structured reporting, and progress tracking influence project outcomes highlighted the practical significance of good work ethics.

The internship also offered lessons in industrial psychology and team dynamics. I observed how different personalities, motivations, and communication styles affect collaboration and decision-making. Conflicts occasionally arose when requirements changed or opinions differed, but I saw that empathy, active listening, and constructive dialogue are key to resolving issues and maintaining a healthy work environment.

Additionally, the experience helped me appreciate organizational hierarchy and the role of leadership in shaping workplace behavior. I realized that supervisors must balance guidance with flexibility, and that employees respond best when they feel valued, understood, and empowered to contribute. This understanding deepened my awareness of how motivation, job satisfaction, and psychological well-being influence productivity.

4.7 Developing My Entrepreneurship Skills

My internship at SSGI provided valuable exposure to entrepreneurial thinking, even within a structured organizational environment. Working on the Internship Management System project helped me understand how innovation, initiative, and problem-solving are central to creating solutions that deliver real value.

I learned to identify opportunities and anticipate needs. For instance, developing a system to manage internship applications and approvals showed me how practical solutions can improve efficiency and reduce manual effort—an entrepreneurial mindset applied within an organizational context. It also taught me to think strategically, considering not only immediate technical requirements but also the long-term usability and scalability of the system.

The experience emphasized the importance of self-learning and adaptability, which are essential traits for any entrepreneur. Having to quickly master Python/Django and PostgreSQL under tight deadlines showed me that taking initiative and learning independently can turn challenges into opportunities.

Furthermore, interacting with stakeholders and understanding their pain points highlighted the value of user-centered thinking, a critical aspect of entrepreneurship. Solutions must meet actual needs while remaining feasible and sustainable, which requires creativity, problem-solving, and foresight.

Overall, my internship strengthened my ability to think innovatively, take initiative, and approach challenges with an entrepreneurial mindset. These skills will not only benefit my future career but also equip me with the confidence and vision needed for potential entrepreneurial ventures.

4.8 Improving My Interpersonal Communication Skills

One of the most valuable lessons from my internship at SSGI was the importance of strong interpersonal communication. Working in the 3T's Department required frequent interaction with team members, supervisors, and other stakeholders, both technical and non-technical.

I learned how to communicate complex technical ideas in simple, understandable terms, ensuring that everyone, regardless of their background, could follow the discussion and

contribute effectively. Preparing weekly presentations reinforced this skill, as I had to summarize developments, explain challenges, and propose solutions clearly and concisely.

The internship also taught me the value of active listening and empathy. Understanding the perspectives and concerns of both supervisors and team members helped me address issues efficiently and prevent misunderstandings. I realized that effective communication is not only about speaking clearly but also about listening, observing, and responding appropriately.

Additionally, collaborating with diverse personalities improved my negotiation and conflict-resolution skills. Whether handling disagreements about system requirements or coordinating tasks among team members, I learned to communicate respectfully, find common ground, and maintain a positive working environment.

This experience enhanced my ability to express ideas clearly, listen effectively, and build strong professional relationships, skills that are essential for any career and particularly valuable in team-based, dynamic workplaces.

CHAPTER FIVE

5.1 General Conclusion and Recommendations for the Company

5.1.1 Conclusion

My internship at the Space Science and Geospatial Institute (SSGI) has been one of the most enriching experiences of my academic journey. At the outset, I was hesitant about the mandatory internship required by AAU. I felt uncomfortable and unsure about stepping into a professional setting, and I questioned the value of this requirement. However, over the course of my time at SSGI, I have completely changed my perspective. The internship allowed me to bridge the gap between theoretical knowledge and practical application, offering insights and skills that simply cannot be gained in a classroom. I now strongly recommend that every student take the opportunity to participate in an internship, as the experience gained is invaluable.

This experience provided hands-on experience with real-world software development, from frontend design to backend development and database management. I learned to work with Python/Django and PostgreSQL, technologies that were entirely new to me, and successfully applied them to create a fully functional system ready for deployment. Beyond technical skills, I gained a deeper understanding of project planning, requirement gathering, and problem-solving under real constraints. Navigating challenges such as changing stakeholder requirements and tight deadlines taught me adaptability and resilience, qualities essential for any professional career.

Additionally, the internship strengthened my soft skills. I learned to communicate effectively with both technical and non-technical staff, collaborate with a diverse team, and

present progress in clear, structured ways. I also observed first-hand how professional ethics, accountability, and discipline influence organizational success. These lessons went far beyond technical knowledge, offering me insights into leadership, teamwork, and the practical dynamics of a professional workplace.

Ultimately, this internship has been transformative. I now see the immense value of practical exposure, mentorship, and professional engagement. The experience has boosted my confidence, expanded my skill set, and prepared me for future challenges in both academic and professional contexts. In conclusion, while I was initially reluctant, I now fully appreciate the necessity and benefits of internships and consider this experience priceless for anyone pursuing a degree and preparing for a career.

5.1.2 Recommendations for the Company

Based on my internship experience and the challenges I encountered, I have several recommendations that could help the 3T's Department, and SSGI as a whole, improve its processes and make projects like ours even more effective in the future.

1. **Clear Requirement Definition Before Development**
   I recommend that the department adopt a structured requirement gathering phase before development begins. This should include written documentation, approval from all stakeholders, and a clear scope definition. Having everyone agree and sign off on requirements at the start will minimize confusion later on.
2. **Better Communication Between Technical and Non-Technical Staff**
   Misalignment often arose because technical designers and non-technical supervisors were not "speaking the same language." I recommend implementing regular workshops or briefing sessions where technical ideas are explained in simple terms, possibly with the help of prototypes, visuals, and flowcharts, to ensure that both sides are aligned.
3. **Change Management Process**
   Since requirements may still evolve, I recommend introducing a formal change management process. Any new request should be documented, reviewed for feasibility, and prioritized based on urgency and resources. This will prevent unnecessary backtracking and ensure that only the most valuable changes are implemented.
4. **Encouraging Team Collaboration and Soft Skills Development**
   While technical skills are important, I realized that teamwork and communication are equally vital. I recommend that the department encourage team-building activities, workshops on communication, and collaboration tools to help interns and staff work better together.

5. **Improved Infrastructure**
   On a smaller but practical note, infrastructure improvements—such as repairing the elevator in the building or improving transport accessibility—would make daily work more convenient. While this may not directly affect project outcomes, it contributes to overall intern and staff satisfaction.

In conclusion, I believe that by addressing these areas—especially requirement clarity, communication, and change management—the department can improve efficiency, reduce frustrations, and achieve even greater success in future projects.

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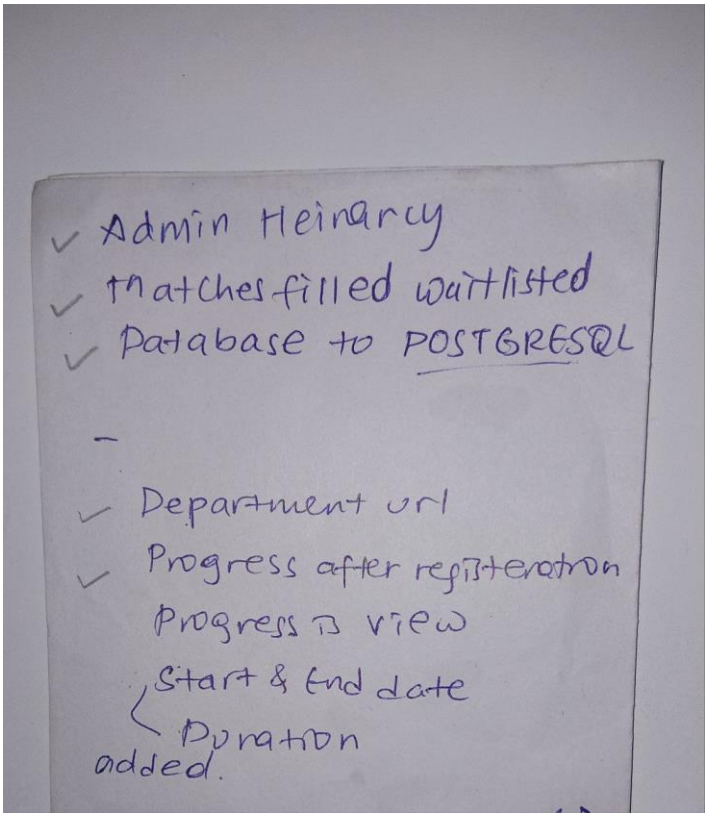
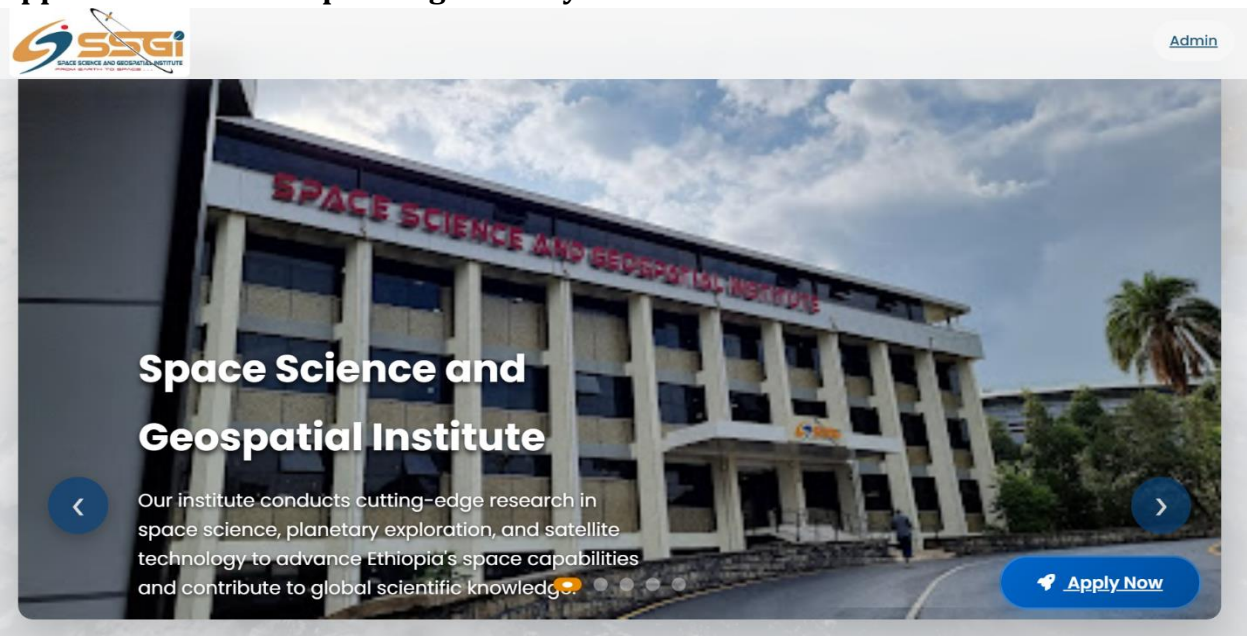
Appendices

Appendix A: Weekly Progress Report Samples -FIGURE 13



Appendix B: Internship Management System Screenshots – FIGURE 14 to 24



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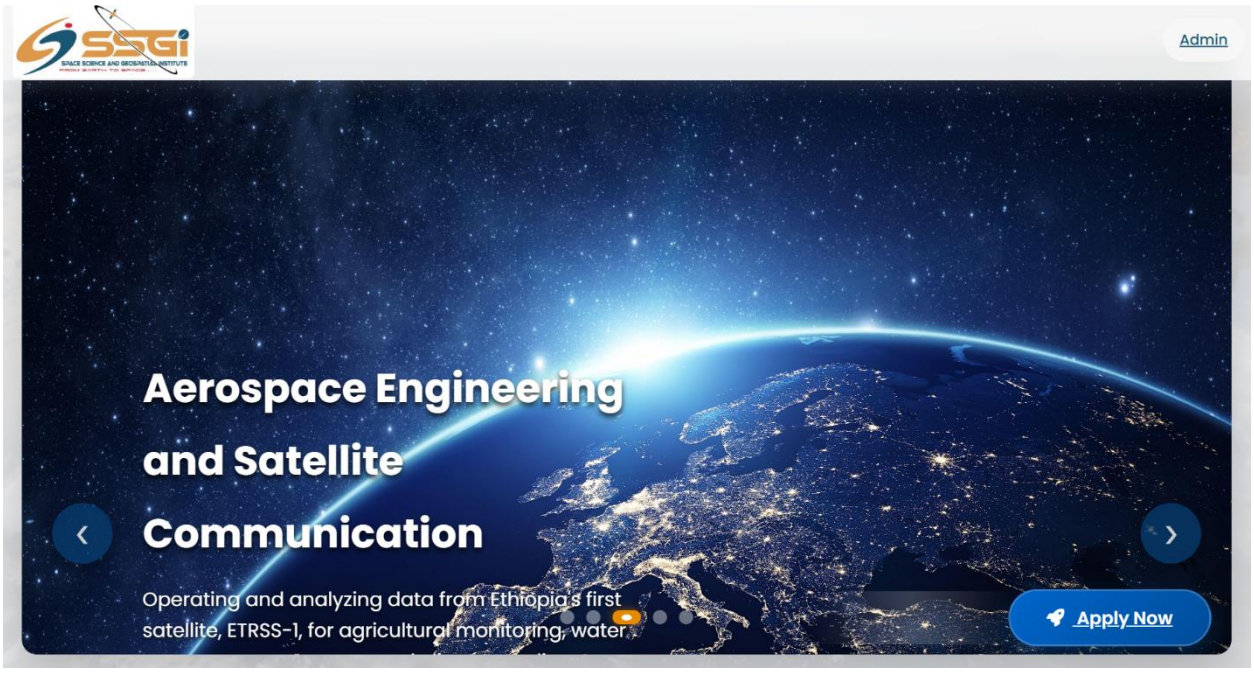
Admin
Aerospace Engineering and Satellite Communication
Operating and analyzing data from Ethiopia's first satellite, ETRSS-1, for agricultural monitoring, water
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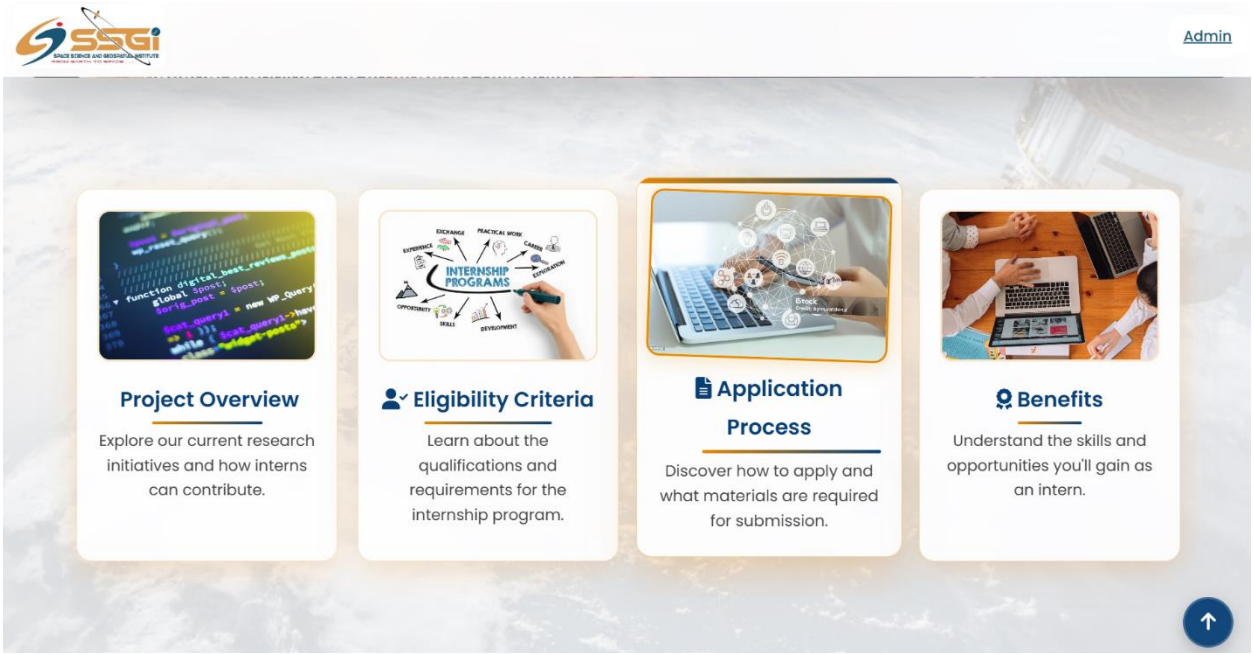
Admin
Project Overview
Explore our current research initiatives and how interns can contribute.
Eligibility Criteria
Learn about the qualifications and requirements for the internship program.
Application Process
Discover how to apply and what materials are required for submission.
Benefits
Understand the skills and opportunities you'll gain as an intern.

STUDENT APPLICATION PAGE

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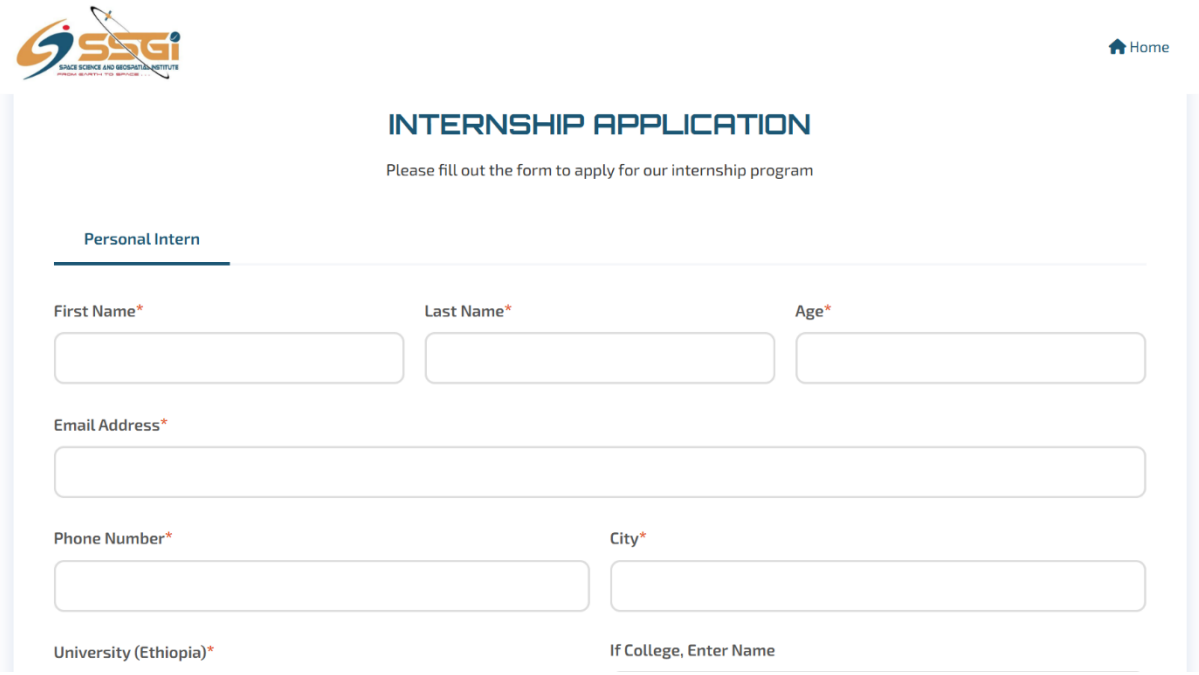
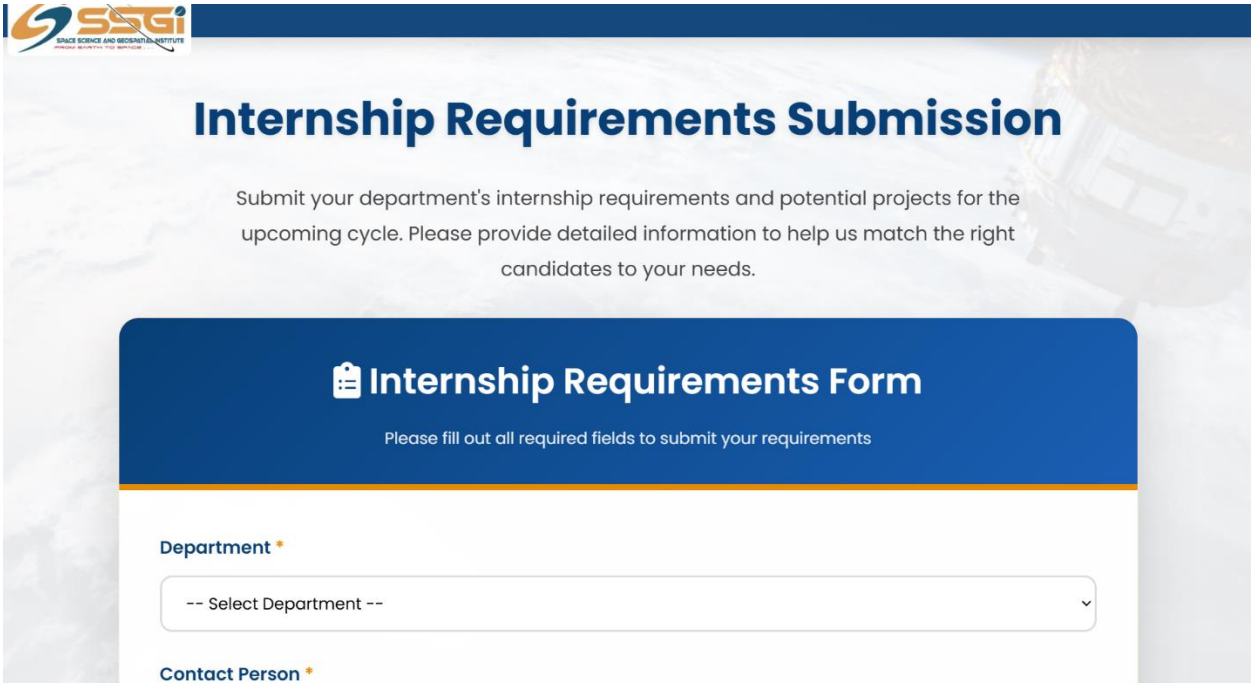
STUDENT APPLICATION FORM



DEPARTMENT REQUIREMENT SUBMISSION PAGE



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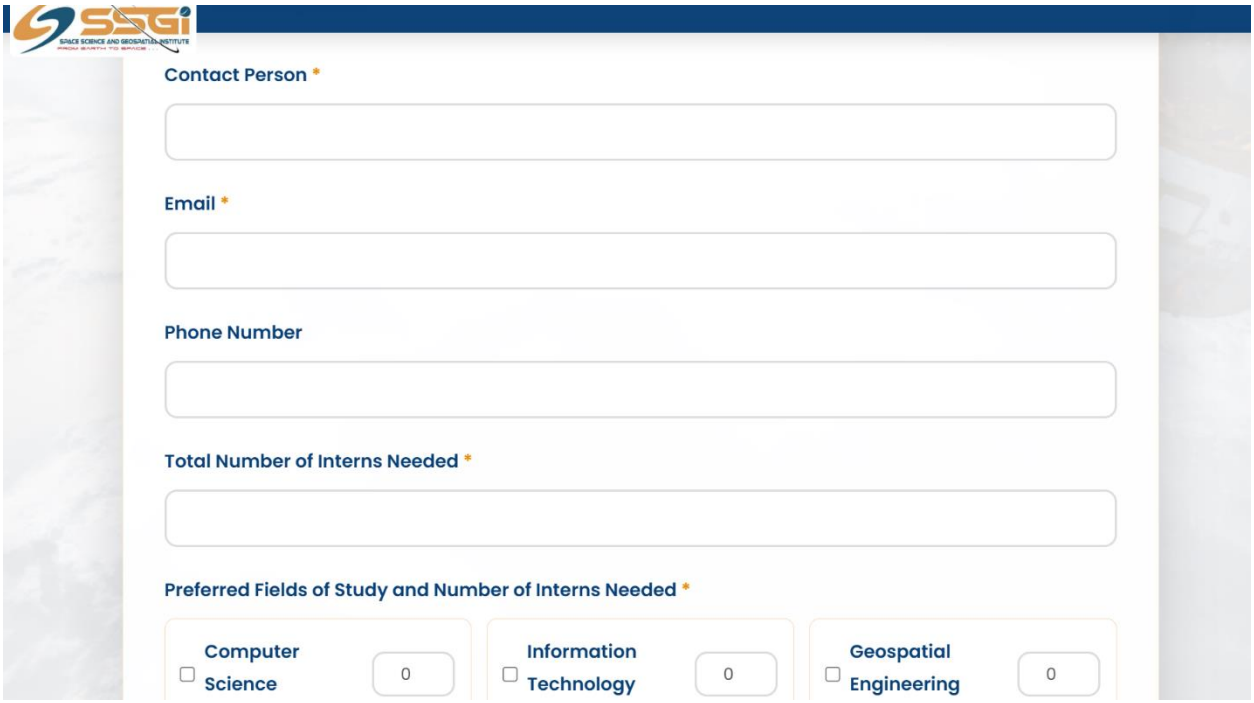
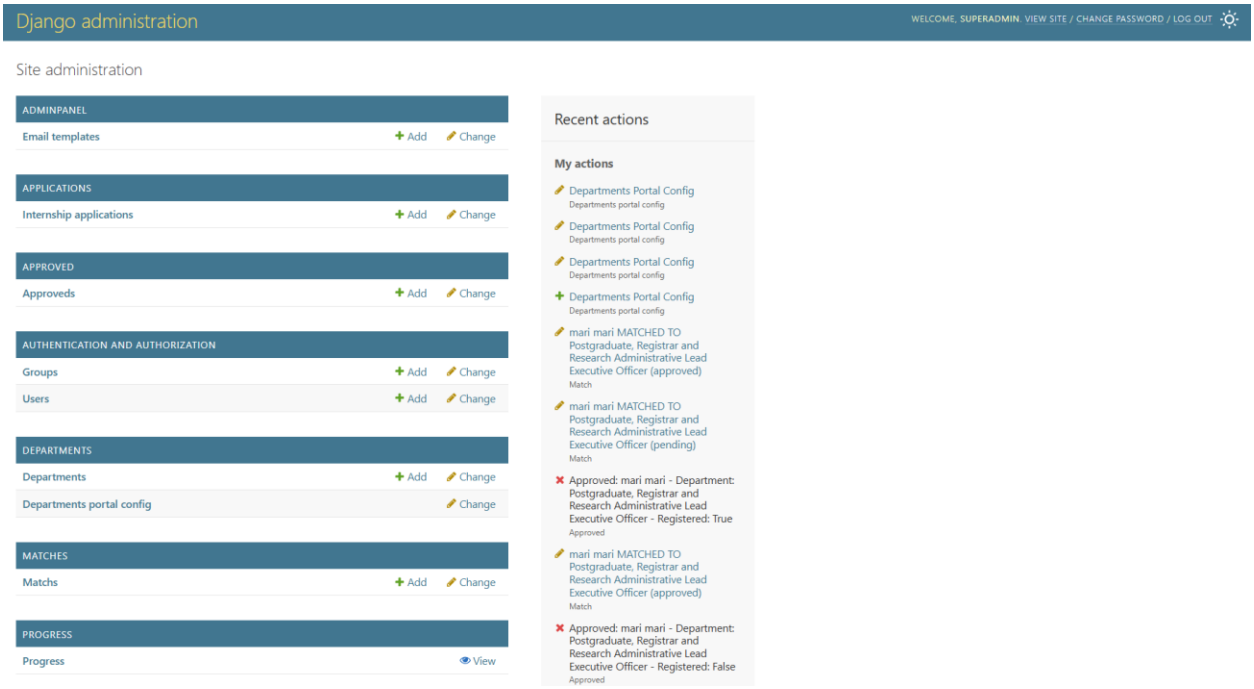


ADMIN DASHBOARD AFTER LOGIN



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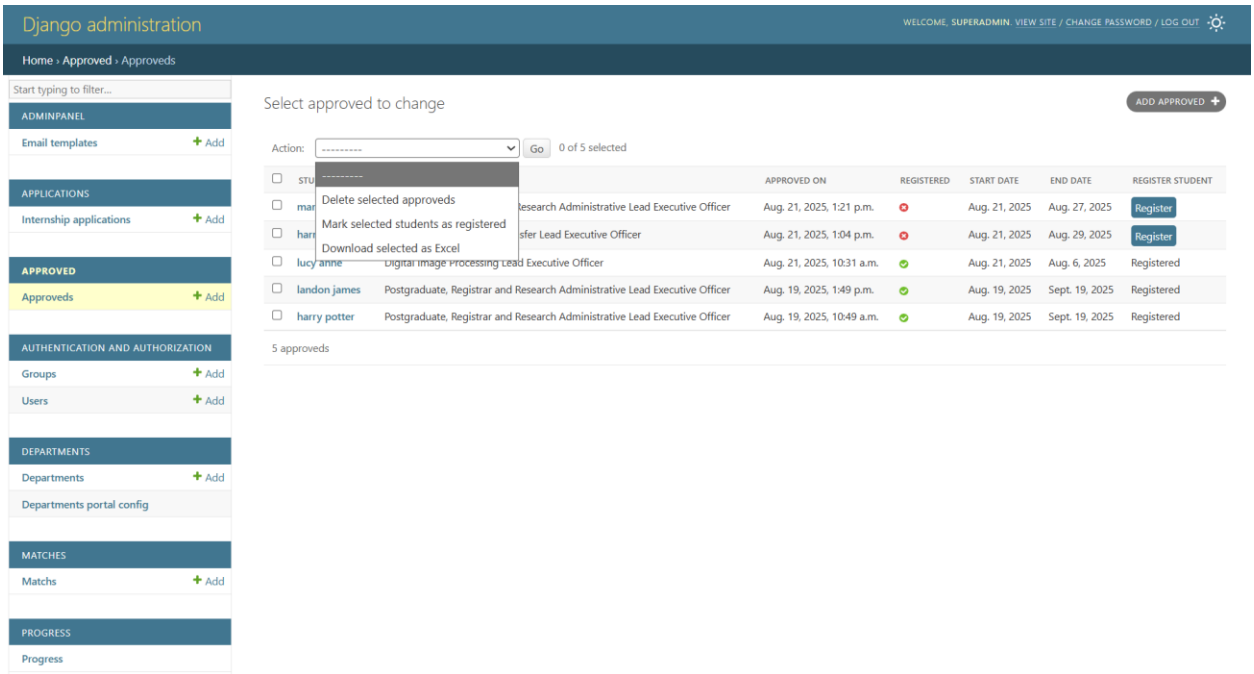
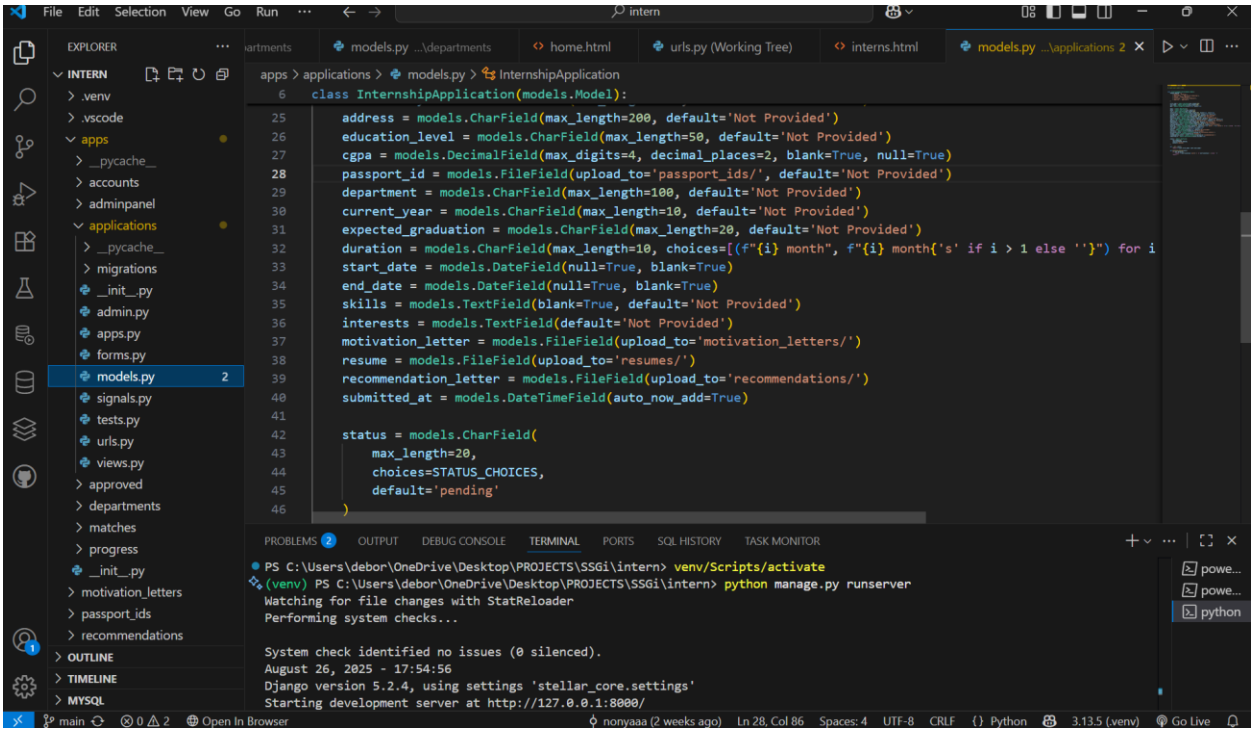
APPROVED STUDENTS TABLE ON THE ADMIN DASHBOARD



BACKEND AND DATABASE SCREENSHOTS



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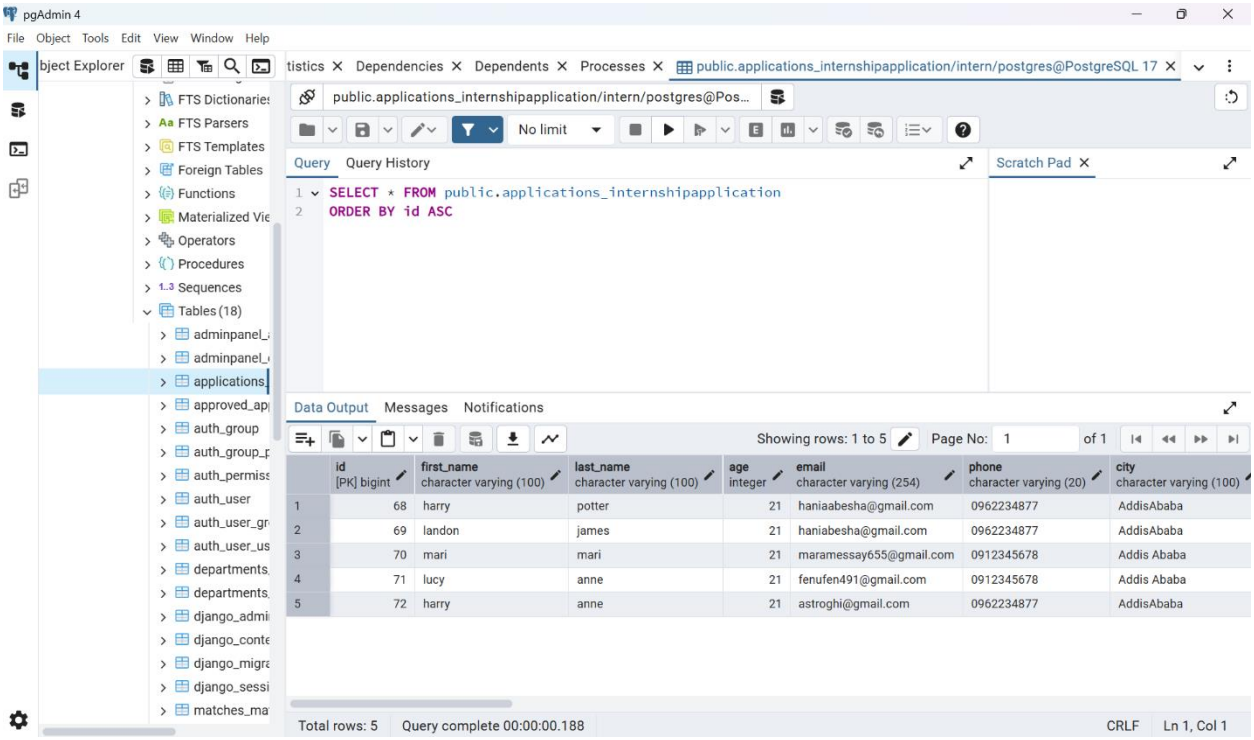

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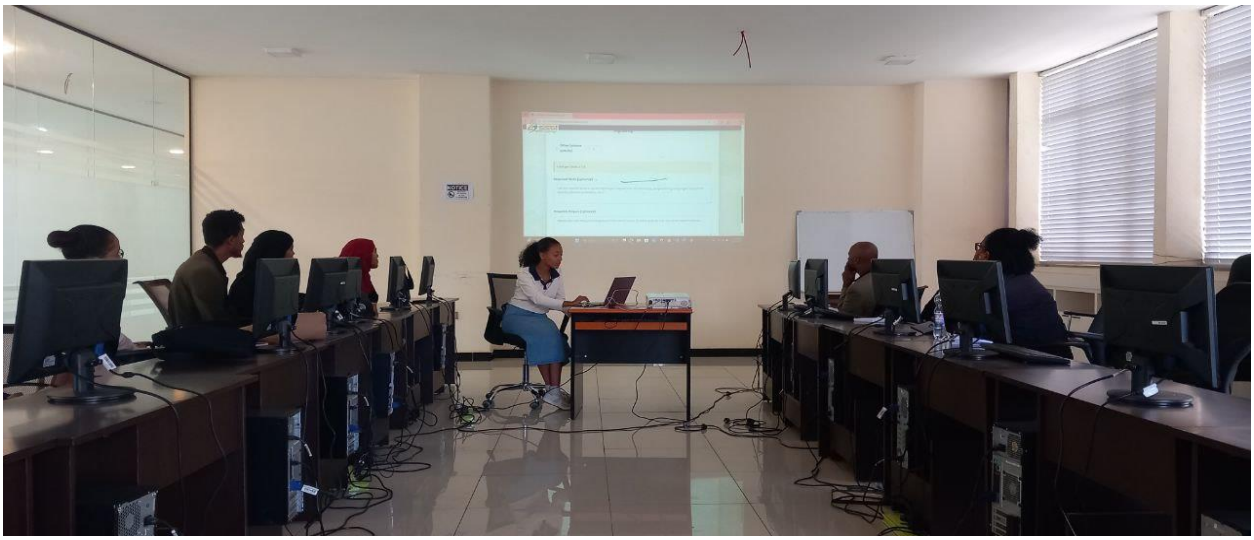
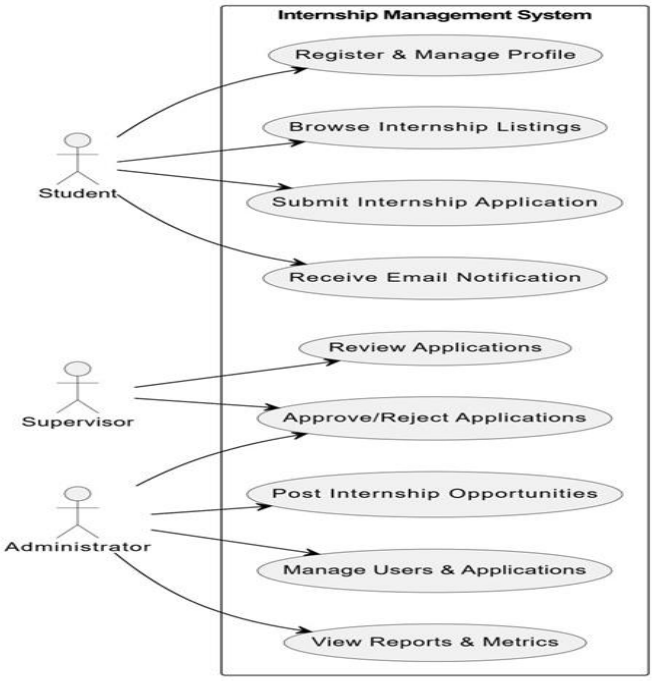
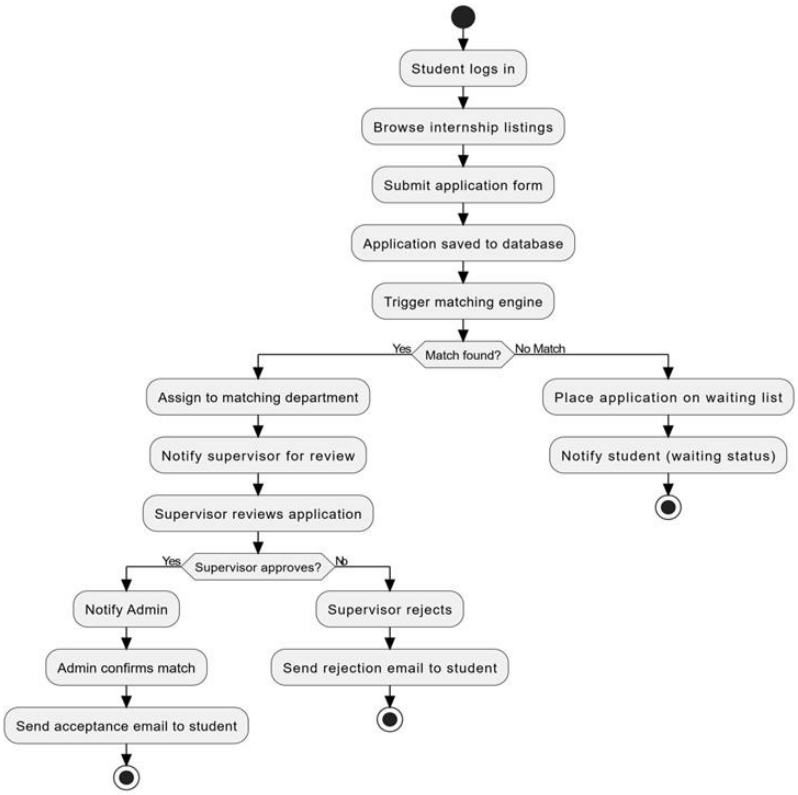
69	landon	james	21	haniabesha@gmail.com	0962234877	AddisAbaba
70	mari	mari	21	maramessay655@gmail.com	0912345678	Addis Ababa
71	lucy	anne	21	fenufen491@gmail.com	0912345678	Addis Ababa
72	harry	anne	21	astroghi@gmail.com	0962234877	AddisAbaba

Appendix C: Presentations – FIGURE 25 to 27



Appendix D: DIAGRAMS USED – FIGURE 28 to 29





Final Note:

I truly loved my internship at SSGI, it was an incredibly insightful and enriching experience that taught me so much, both professionally and personally. I am deeply grateful to the institute for giving me this opportunity, to my supportive family, my university, my internship advisor, and especially the amazing team I had the privilege to work with. The friendships, guidance, and encouragement I received made this journey unforgettable, and I will always cherish the lessons and memories from this experience.

Thank you all.